Venous Oxygen Saturation

Authors

Sanjana Chetana Shanmukhappa[1]; Srivatsa Lokeshwaran[2].

Affiliations

[1] Aster CMI Hospital
[2] Aster CMI Hospital

Last Update: October 24, 2022.

Introduction

Venous oxygen saturation (SvO2) is a measure of the oxygen content of the blood returning to the right side of the heart after perfusing the entire body. When the oxygen supply is insufficient to meet the metabolic demands of the tissues, an abnormal SvO2 ensues and reflects an inadequacy in the systemic oxygenation. SvO2 is, therefore, dependent on oxygen delivery and oxygen extraction.

Venous oximetry is used in certain clinical settings of hemodynamic instability, such as in critical illnesses, perioperative periods of major surgeries, heart failure, and sepsis. The venous blood in the pulmonary artery represents oxygen extraction in the whole body and is called mixed venous oxygen saturation (SmvO2). It was measured using a pulmonary artery catheter (PAC). A second, less invasive method of measuring SvO2 is via a central venous catheter (CVC) positioned in the superior vena cava and is called the central venous oxygen saturation (ScvO2).

Specimen Collection

Due to its invasive nature, the use of PACs is limited to conditions where the hemodynamic status of a critically ill patient cannot be assessed accurately by other means.[1]

Informed consent should be taken prior to the PAC insertion. The equipment should be calibrated, set up, and checked before insertion. Sterile technique and barrier precautions must be maintained during the procedure. Once the patient is appropriately positioned, the PAC is inserted into one of the central veins, such as the subclavian, internal jugular, or femoral veins. This is done under either ultrasound or fluoroscopic guidance. The PAC is passed through the superior or inferior vena cava into the right heart and advanced into the pulmonary artery (PA). Once the required position is reached, a chest radiograph is obtained for confirmation and to rule out any adverse events.

Hemodynamic parameters measured by a PAC includes pressures in the right atrium (RA), right ventricle (RV), PA, pulmonary capillary wedge pressure (PCWP), as well as the cardiac output (CO). SmvO2 is measured by slow aspiration of a mixed venous blood sample drawn from a distal port of the PAC leading into the PA.[2]

In contrast to PAC, CVCs are more commonly used, associated with lesser complications, and cost-effective.[3] Additionally, most critically ill patients already have a central line in place for hemodynamic monitoring, volume resuscitation, nutritional support, and more.

As with the placement of PAC, informed consent is obtained before the procedure, and the technique is carried out under sterile conditions. The equipment and devices needed are obtained, and the patient is placed in an anatomically advantageous position (Trendelenberg position for subclavian/internal jugular vein access). Pre and peri-procedural ultrasound is used to define anatomy, to reduce the time for venous access and the risk of complications. The CVC is introduced into one of the central veins (decided based on the patient condition and comfort of the practitioner), and the tip is positioned to lie in the lower superior vena cava or outside the right atrium.[4] The catheter placement is confirmed by one of the following methods: chest radiography, fluoroscopy, ultrasound, or transesophageal echocardiography. A venous blood sample is drawn for ScvO2 measurement.

Continuous vs. Intermittent Measurement of Venous Saturation

ScvO2 or SmvO2 can be measured by drawing blood from the distal line of CVC or PAC for blood gas analysis. It can also be measured continuously using a fibreoptic catheter that uses reflection spectrophotometry. The saturation value is displayed on an oximetry monitor and updated every 2 seconds. Thus, this provides up-to-date real-time measurement of the venous oxygen saturation, and fluctuations can be closely monitored in critically ill patients. While the continuous venous oximetry is more expensive, repeated blood draws for blood gas analysis in unstable patients leads to blood loss and adds up to the cost. A pilot study comparing the two in sepsis concluded that intermittent measurement was not inferior to continuous monitoring when delivered with the first 6 hours of treatment.[5] These factors should be considered when deciding the method of venous oximetry.

Procedures

Procedure for pulmonary artery catheter placement (after the consent, equipment arrangement, preparation, and positioning of the patient):

- Using a high-frequency ultrasound probe, the desired vein is identified. The surrounding skin is anesthetized using 1% lignocaine.
- Access to the vein is obtained using an appropriate gauge needle. Once access is confirmed, the syringe is removed, leaving the needle in the vessel and a guidewire is inserted through the needle into the vein.
- The needle is then removed leaving the guidewire in place. A small incision is made at the entry of the guidewire.
- An introducer sheath is inserted, threaded into the vein and the guidewire is removed.
- If the PAC is intended to be left in place for long durations, a sterile sleeve is placed over the catheter prior to its insertion to allow for some adjustments in the position of the catheter at a later time (whilst maintaining sterility).
- The PAC is inserted through the introducer sheath and advanced into the RA. Once it is inserted up to the 20cm mark, the balloon of the catheter is free of the sheath and is inflated with 1.5ml of air.
- The catheter tip is advanced slowly through the sheath while observing for the characteristic waveforms and pressures from the RA->RV->PA->PCWP. All pressures are measured at the end of expiration.
- When the PCWP is obtained, the balloon is deflated, and the catheter is withdrawn slightly to observe the PA trace reappear.
- A chest radiograph is obtained to ensure the correct placement of the catheter.

Procedure for central venous catheter insertion (after the consent, equipment arrangement, preparation, and positioning of the patient):

- Using ultrasound guidance the desired vein is identified and confirmed. The skin is infiltrated with 1% lignocaine to anesthetize cutaneously and subcutaneously.
- Under dynamic ultrasound visualization, the introducer needle is used to cannulate the vein.
- This is followed by the insertion of a guidewire through the needle.
- While holding the inserted wire steady inside the vein, the needle is removed.
- Advance the tissue dilator over the guidewire to dilate the soft tissues and aid the passage of the central venous catheter.
- The dilator is removed and the catheter is threaded over the guidewire.
- Once the catheter is in place, the guidewire is removed.
- Blood is aspirated from each port and flushed with saline.
- The catheter is sutured in place and a sterile dressing is applied.
- The position of the tip of the catheter is confirmed using chest radiography (for subclavian/internal jugular vein).

Indications

Venous oxygen saturation has diagnostic, therapeutic as well as prognostic value in critically ill patients. As a diagnostic, derangement in venous oxygen saturation can ascertain the underlying etiology. The oxygen consumption by the tissues by the whole body (VO2) is usually independent of oxygen delivery (DO2). This is because a decrease in DO2 is compensated by an increase in VO2, thereby preventing tissue hypoxia. However, when a 'critical' DO2 is reached, and no further oxygen can be extracted, tissue hypoxia and lactic acidosis sets in. These changes are reflected in venous oximetry and must be interpreted correctly in conjunction with other hemodynamic parameters (CO, arterial oxygen content).

The cardiovascular response to an increase in oxygen requirement and VO2 is an increase in arterial oxygen content or an increase in CO. When these compensatory mechanisms fail to occur, and DO2 drops, a decrease in SvO2 is seen. SvO2 also drops if the compensatory mechanisms are insufficient to meet the metabolic demands of the tissues. SvO2 can drop to as low as 30-50% before tissue oxygen extraction is exhausted, and there is an onset of anaerobic metabolism. Venous oximetry is a therapeutic parameter for 'early goal-directed therapy' (EGDT) and is useful in critical conditions (such as sepsis) to monitor changes in SvO2 following interventions.

The following is a summary of the clinical conditions underlying SvO2 alterations:[6]

Increase in SvO2

1. Increase in DO2

- Increase in CO
- Increase in arterial oxygen content

2. Decrease in VO2 VO2 = oxygen consumption by the tissue of the entire body

- Hypothermia
- Sedation
- Analgesia
- Mechanical ventilation
- Sepsis
- Decreased oxygen extraction (cell death, A-V shunting)

Decrease in SvO2

1. Decrease in DO2

- Decrease in CO
- Decrease in arterial oxygen content (hypoxia, anemia)

2. Increase in VO2 VO2 = oxygen consumption by the tissue of the entire body

- Pain
- Stress
- Shivering
- Hyperthermia
- Infection
- Seizures

Potential Diagnosis

1. Cardiac Failure and Cardiac Arrest

SvO2 drops in patients with left ventricular failure or cardiogenic shock. Fall in CO results in changes in DO2 and a subsequent increase in VO2, ultimately leading to inverse changes in SvO2. Patients with chronic heart failure may be adapted to a low SvO2 (30 to 40%) due to chronic tissue hypoxia. An acute drop in SvO2 is an indication of cardiac deterioration. On the other hand, SvO2 improving (to >72%) following cardiopulmonary resuscitation is a marker for the return of spontaneous circulation. Thus, in such settings, it is useful to monitor SvO2.

2. Severe Trauma

SvO2 (<65%) after trauma predicts a loss of blood and a need for blood transfusion. Additionally, a low SvO2 in the first 24 hours following a major trauma/head injury is a prognostic indicator of higher mortality and longer hospital stay.[7]

3. Major Surgery

Monitoring SvO2 in high-risk surgery has been shown to be beneficial for the early detection of hemodynamic deterioration. This indicates an inability of the heart to increase the CO in the setting of occult blood loss or anemia in patients with cardiac dysfunction. As with trauma, a low SvO2 indicates a postoperative increase in morbidity and mortality. A decrease in SvO2 when weaning a patient off of mechanical ventilation indicates a failure of extubation. The period of intraoperative derangement of SvO2 is a predictor of postoperative complications.

Anesthesia, analgesia, sedation, mechanical ventilation, blood transfusion, and intravenous fluid administration are all routinely utilized perioperative interventions that influence venous oxygen saturation. Studies have shown that a goal-directed approach intraoperatively reduces complications, hospital length of stay, and mortality postoperatively. Strict adherence to predefined hemodynamic goals (cardiac index, SvO2, oxygen extraction ratio, urine output) has demonstrated a lesser occurrence of organ dysfunction and an overall improved outcome.[8] However, owing to the risks involved when repeated interventions are done to attain therapeutic goals, this strategy is not applied to all patients.

4. Sepsis and Septic Shock

Sepsis is characterized by myocardial depression, hypovolemia, hypotension, and organ dysfunction. Septic shock is defined as sepsis with global tissue hypoxia and hypotension not correct by fluid resuscitation. 'Early goal-directed therapy' (EGDT) developed by Rivers *et al.* in 2001 is an accepted approach to sepsis and septic shock management.[9] It involves early recognition of sepsis and its aggressive management in the Emergency Room. The components of EGDT (in addition to standard care) include fluid resuscitation and central venous pressure monitoring, vasopressors and mean arterial pressure, blood transfusion, and SvO2 monitoring. The goal for ScvO2 is >70% or SmvO2 >65%.

While many studies have clearly demonstrated the benefit of SvO2 monitoring for EDGT, several recent randomized trials have failed to show any improvement in survival.[10] These discrepancies are being studied widely, and the reliability of SvO2 is a continued source of much debate.

Normal and Critical Findings

In normal patients, the ScvO2 is 2 to 3% lower than SmvO2 because the lower body extracts less oxygen than the upper body. In non-shock conditions, there is a good correlation between ScvO2 and SmvO2. However, in shock, there are changes in regional blood flow and oxygen extraction. There is a decrease in mesenteric and renal blood supply (with an increase in O2 extraction) and redistribution of blood flow to the GI tract and the brain in the later stages of shock. Thus, the absolute numerical values of ScvO2 and SmvO2 are not comparable. However, the trends in change parallel one another, and therefore the trend of ScvO2 is often used as a substitute for SmvO2. Additionally, it is important to note that factors such as sedation, recent intubation, and position of the tip of the central catheter (closer to the right atrium, more comparable to SmvO2) must be taken into consideration when interpreting the ScvO2 values. In critical illnesses, ScvO2 is estimated to be higher than SmvO2 by 7% +/- 4%.

PACs are preferred over CVCs in selected cases of patients with critical illnesses. It is a valuable diagnostic and hemodynamic monitoring tool that provides data on mixed venous oxygen saturation, cardiac chamber pressures, CO, and more. Its use in cardiac surgery has shown an overall benefit with a decrease in length of hospital stay and reduced cardiopulmonary morbidity. However, there is an increased occurrence of infections. In patients with sepsis, PACs have failed to demonstrate a clear mortality benefit. Due to its disputed efficacy and safety, its use is limited to aid decision-making in severely ill patients and unresponsive to initial resuscitative measures.[11]

Values and interpretation of SmvO2: [12]

1. SmvO2 >75%

Normal oxygen extraction (Indicates adequate O2 supply)**2. 75% >SmvO2 >50%**

Compensatory oxygen consumption (Indicates an increase in O2 demand or a decrease in DO2)**3. 50% >SmvO2 >30%**

Exhaustion of extraction (Indicates the beginning of lactic acidosis as the 'critical DO2' is reached and no more oxygen can be extracted)**4. 30% >SmvO2 >25%**

Severe lactic acidosis**5. SmvO2 <25%**

Cell death

Interfering Factors

Several factors interfere with the readings obtained from a CVC/PAC. These include patient-related factors such as improper positioning, wrong vein selection, patient movements, or the presence of a thrombus. Faulty device calibration and zeroing are technical issues that affect readings. Catheter-related snags include improper placement of the catheter tip or kinking of the catheter.

Complications

Complications Related to PAC Placement

1. Arrhythmias when the catheter is introduced into the cardiac chambers due to wall irritation. 2. Knotting of the catheter inside the cardiac chambers identified on chest radiography. 3. Misplacement due to the looping of the catheter in the right atrium or ventricle. 4. Valve rupture or endocardial injury.5. Pulmonary artery perforation, a grave complication that occurs due to balloon overinflation or perforation of the catheter tip. It presents as hemoptysis and requires urgent management. 6. Pulmonary infarction occurs if the catheter is left wedged for a prolonged duration of time or migration of the tip to distal branches. 7. Thromboembolism (less common now because of the use of heparin bonded catheters). 8. Air embolism is caused by open infusion ports and entry of air into the venous circulation. 9. Catheter-related bloodstream infections.

Complications Related to CVC Placement

1. Cardiovascular complications such as arrhythmias, venous or arterial vessel injury or rupture, bleeding, and hematoma formation. 2. Pneumothorax is more frequent with subclavian vein catheterization. 3. Venous air embolism, a fatal complication that occurs when the needle or catheter is left open. 4. Indwelling catheter-related bloodstream infections

Patient Safety and Education

Prior to catheter placement, you are counseled and explained about the procedure, its need, benefits, and potential risks. At this time, it is important that you or your caregiver provide information that may affect the line placement, such as allergies (to dyes/antibiotics/lignocaine), previous line placements, previous line infections, or other details you deem relevant. As per your condition, before line insertion, necessary investigations may be ordered or blood samples drawn to check for bleeding disorders. You are asked not to eat or drink for 6 hours prior to the procedure.

If you are required to be sent home with the central venous catheter in place at the time of discharge, a nurse or health care provider will teach you how to take care of the line. It is important to keep the line clean (by practicing hand hygiene/using alcohol-based hand rubs before handling the line), dry (by taking only sponge baths if needed), and avoiding damage to the catheter by tugging or using sharp objects. Remember to seek medical care if you notice any signs of infection (redness, swelling, pus formation at the site of insertion or developing fever/chills), difficulty breathing, and if the catheter is damaged or falls out.

Clinical Significance

Venous oximetry is the key that provides invaluable insight into the adequacy of the cardiopulmonary system. It yields an estimation of the oxygen supply-demand ratio in the human body during critical illnesses. It assists in the early recognition and management of tissue hypoxia. Venous oxygen saturation levels should be used in conjunction with other vitals hemodynamic parameters (cardiac output, heart rate, blood pressure), serum lactate levels, and urine output to guide the patient's treatment plans. However, much remains to be understood about translating the information provided by it into therapeutic protocols for beneficial outcomes.